QRU • GENERAL

PRINTABLE PDF

Sports Multiple 36 Kids — Printable PDF

A For Children learning product

Foundational • General public

Sports Multiple 36 Kids - Printable PDF

QRU PRESS™

Edition 1

TREASURE STANDARD(TM) CERTIFIED

Sports Multiple 36 Kids - Printable PDF

Manufactured by QRU Factory(TM). QRU simplifies the path to understanding the truth.

QRU Sports Intelligence Lab(TM)

Sports Multiplication for Kids

36 Brain Athlete Training Challenges

Audience: General Public
Learning Level: Introductory
Purpose: Help children build math understanding, confidence, reasoning, and persistence through sports-based learning.

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1. Big Idea

Children do not need more information first.
They need stronger understanding systems.

QRU Sports Intelligence Lab(TM) turns multiplication into a sports training experience. Instead of treating math as only memorization, this approach connects multiplication to:

- Sports
- Practice
- Strategy
- Teamwork
- Decision-making
- Confidence-building
- Reflection

In this lab, the learner becomes a Brain Athlete.

A Brain Athlete trains the mind the way an athlete trains the body.

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2. What Is Multiplication?

Multiplication means many equal groups combined.

Example:

If a basketball player makes 8 free throws each practice and practices 5 times, then:

$8 \times 5 = 40$

That means:

5 groups of 8 = 40 total free throws

3. Everyday Analogy

Learning math is like practicing a sport:

One missed shot does not mean you are bad at basketball.
One wrong answer does not mean you are bad at math.
Both are training reps that help you improve.

Mistakes are not proof that you cannot learn.
Mistakes are part of learning.

4. The Brain Athlete Mindset

A Brain Athlete does not ask only:

"Did I get it right?"

A Brain Athlete also asks:

- What strategy did I use?
- What pattern did I notice?
- What mistake can help me improve?
- Can I explain my thinking?
- How can I train again tomorrow?

5. Coach Notes for Adults

Parents, teachers, and caregivers are coaches.

The goal is not perfection.
The goal is helping the child discover:

"I can learn difficult things."

Encourage the child to:

- Try before being shown the answer

- Explain their strategy
- Use drawings, objects, or skip-counting
- Learn from mistakes
- Celebrate improvement
- Practice over time

Avoid making multiplication only about speed.
Speed can come later. Understanding comes first.

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6. QRU Training Cycle

Use this cycle for each problem:

1. Question - What is being asked?
2. Understanding - What do the numbers mean?
3. Practice - Solve the problem.
4. Connection - How does this connect to sports or real life?
5. Application - Can you create a similar problem?
6. Reflection - What did you learn?
7. Growth - What will you practice next?

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7. Vocabulary Decoder(TM)

Multiplication

Multi = many
Plication = groups or folds

Multiplication means many groups combined.

Product

The answer to a multiplication problem.

Example:

$6 \times 4 = 24$

The product is 24.

Factor

A number being multiplied.

Example:

6 × 4 = 24

The factors are 6 and 4.

Intelligence

The ability to learn, understand, adapt, and solve problems.

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8. Sports Examples

Football Example

A quarterback completes 5 passes each quarter.
There are 4 quarters.

5 × 4 = 20

The quarterback completes 20 passes.

Basketball Example

A player makes 8 free throws each practice.
The player practices 5 times.

8 × 5 = 40

The player makes 40 free throws.

Soccer Example

A soccer player takes 6 shots each game.
The player plays 3 games.

6 × 3 = 18

The player takes 18 shots.

Baseball Example

A team scores 4 runs each inning for 2 innings.

4 × 2 = 8

The team scores 8 runs.

9. What Multiplication Is Not

Multiplication is not:

- Memorizing without understanding
- Racing against others to prove intelligence
- A measure of a child's worth
- Something only "math people" can learn

A child who struggles may need:

- More practice
- A different explanation
- A visual model
- A real-world connection
- More time
- Encouragement

10. The 36 Brain Athlete Training Challenges

Use these challenges as a printable practice set.

For each challenge:

1. Read the story.
2. Write the multiplication equation.
3. Solve the product.
4. Explain your strategy.

Level 1: Rookie Warm-Up

1. Basketball Free Throws

A player makes 3 free throws in each round.
There are 4 rounds.

Equation: × =

2. Soccer Passes

A soccer player makes 5 passes each drill.
There are 2 drills.

Equation: × =

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3. Football Catches

A receiver catches 4 passes each quarter.
There are 3 quarters.

Equation: × =

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4. Baseball Hits

A player gets 2 hits each game.
The player plays 6 games.

Equation: × =

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5. Swimming Laps

A swimmer swims 5 laps each set.
There are 3 sets.

Equation: × =

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6. Tennis Serves

A tennis player practices 6 serves each round.
There are 2 rounds.

Equation: × =

Level 2: Practice Field

7. Basketball Shots

A player takes 7 shots each practice.
There are 3 practices.

Equation: × =

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8. Soccer Goals

A team scores 2 goals each game.
They play 5 games.

Equation: × =

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9. Football Touchdowns

A team scores 3 touchdowns each game.
They play 4 games.

Equation: × =

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10. Baseball Innings

A team scores 4 runs each inning.
They score for 3 innings.

Equation: × =

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11. Volleyball Blocks

A player makes 5 blocks each match.
There are 4 matches.

Equation: × =

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12. Track Sprints

A runner completes 6 sprints each workout.
There are 3 workouts.

Equation: × =

Level 3: Game Strategy

13. Basketball Quarters

A player scores 8 points each quarter.
There are 4 quarters.

Equation: × =

14. Soccer Training Cones

A coach sets up 9 cones in each row.
There are 3 rows.

Equation: × =

15. Football Practice Plays

A team runs 7 plays each drill.
There are 4 drills.

Equation: × =

16. Baseball Batting Practice

A player takes 8 swings each round.
There are 5 rounds.

Equation: × =

17. Swimming Relay

Each swimmer swims 4 laps.
There are 6 swimmers.

Equation: × =

18. Tennis Rally

Two players hit the ball back and forth 10 times each rally.
They complete 3 rallies.

Equation: × =

Level 4: Team Challenge

19. Basketball Team Points

Each player scores 6 points.
There are 5 players.

Equation: × =

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20. Soccer Team Drills

Each player completes 4 drills.
There are 11 players.

Equation: × =

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21. Football Yard Gains

A running back gains 9 yards each carry.
The player has 4 carries.

Equation: × =

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22. Baseball Cards

A child collects 7 baseball cards each week.
They collect for 6 weeks.

Equation: × =

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23. Gymnastics Routines

A gymnast practices 5 routines each day.
They practice for 7 days.

Equation: × =

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24. Hockey Shots

A hockey player takes 8 shots each game.
They play 6 games.

Equation: × =

Level 5: Pattern Power

25. Basketball Rebounds

A player gets 9 rebounds each game.
They play 5 games.

Equation: × =

26. Soccer Jerseys

There are 10 jerseys in each box.
The coach opens 6 boxes.

Equation: × =

27. Football Helmets

There are 11 helmets on each rack.
There are 3 racks.

Equation: × =

28. Baseball Bats

There are 12 bats in each bag.
There are 4 bags.

Equation: × =

29. Swimming Practice Minutes

A swimmer practices 10 minutes each station.
There are 5 stations.

Equation: × =

30. Track Team Runners

There are 8 runners in each group.
There are 7 groups.

Equation: × =

Level 6: Champion Round

31. Basketball Tournament

A team plays 6 games.
They score 12 points in each game.

Equation: × =

32. Soccer Tournament

A player takes 7 shots each game.
They play 8 games.

Equation: × =

33. Football Practice Week

A team runs 9 plays each practice.
They practice 6 times.

Equation: × =

34. Baseball Season

A player gets 3 hits each game.
They play 12 games.

Equation: × =

35. Volleyball Serves

A player practices 11 serves each day.
They practice for 5 days.

Equation: × =

36. All-Star Brain Athlete

A Brain Athlete solves 12 problems each day.
They train for 6 days.

Equation: × =

11. Answer Key

1. $3 \times 4 = 12$
2. $5 \times 2 = 10$
3. $4 \times 3 = 12$
4. $2 \times 6 = 12$
5. $5 \times 3 = 15$
6. $6 \times 2 = 12$
7. $7 \times 3 = 21$
8. $2 \times 5 = 10$
9. $3 \times 4 = 12$
10. $4 \times 3 = 12$
11. $5 \times 4 = 20$
12. $6 \times 3 = 18$
13. $8 \times 4 = 32$
14. $9 \times 3 = 27$
15. $7 \times 4 = 28$
16. $8 \times 5 = 40$
17. $4 \times 6 = 24$
18. $10 \times 3 = 30$
19. $6 \times 5 = 30$
20. $4 \times 11 = 44$
21. $9 \times 4 = 36$
22. $7 \times 6 = 42$
23. $5 \times 7 = 35$
24. $8 \times 6 = 48$
25. $9 \times 5 = 45$
26. $10 \times 6 = 60$
27. $11 \times 3 = 33$
28. $12 \times 4 = 48$
29. $10 \times 5 = 50$
30. $8 \times 7 = 56$
31. $12 \times 6 = 72$
32. $7 \times 8 = 56$
33. $9 \times 6 = 54$
34. $3 \times 12 = 36$
35. $11 \times 5 = 55$
36. $12 \times 6 = 72$

12. Strategy Toolbox

Children may solve multiplication problems using different strategies.

Strategy 1: Equal Groups

Draw groups.

Example:

4 × 3 means 4 groups of 3.

???
???
???
???

Total: 12

Strategy 2: Repeated Addition

Example:

5 × 4

5 + 5 + 5 + 5 = 20

Strategy 3: Skip Counting

Example:

6 × 5

Count by 6 five times:

6, 12, 18, 24, 30

Answer: 30

Strategy 4: Arrays

Arrange numbers in rows and columns.

Example:

3 × 4

? ? ? ?
? ? ? ?
? ? ? ?

3 rows of 4 = 12

Strategy 5: Known Facts

Use facts you already know.

Example:

If you know:

5 × 6 = 30

Then:

6 × 6 = 36

because it is one more group of 6.

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13. Reflection Questions

After completing the challenge set, answer these questions.

1. Which problem felt easiest? Why?
2. Which problem felt hardest? Why?
3. What strategy helped you the most?
4. What mistake taught you something?
5. How did sports help the math make more sense?
6. What multiplication fact do you want to practice next?
7. How are you becoming a stronger Brain Athlete?

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14. Create Your Own Sports Problem

Choose a sport:

- Basketball
- Soccer
- Football
- Baseball
- Tennis
- Swimming
- Track
- Volleyball
- Hockey
- Gymnastics
- Your favorite sport:

Now create your own multiplication story.

Sport:

Story Problem:

Equation: × =

Answer:

My Strategy:

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15. Mini Quiz

Recall Questions

1. What does multiplication mean?
2. What is the answer to a multiplication problem called?
3. What are the numbers being multiplied called?

Application Questions

4. A player scores 4 points each round for 5 rounds. What is the total?
5. A swimmer swims 6 laps each day for 4 days. What is the total?

Reasoning Questions

6. Why is 3 × 5 the same total as 5 × 3?
7. How can drawing groups help you understand multiplication?

Reflection Question

8. What does a Brain Athlete do when a problem feels hard?

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16. Mini Quiz Answer Guide

1. Multiplication means many equal groups combined.
2. The product.
3. Factors.
4. 4 × 5 = 20
5. 6 × 4 = 24
6. Both make 15 because the same total can be arranged as 3 groups of 5 or 5 groups of 3.
7. Drawing groups shows what the numbers mean instead of only memorizing.
8. A Brain Athlete keeps practicing, tries a strategy, asks questions, and learns from mistakes.

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17. Memory Sentences(TM)

Champions train their bodies.
Brain Athletes train their minds.

Understanding creates confidence.

Mistakes are training reps.

Math is not just answers. Math is understanding patterns.

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18. Memorable Takeaway

Multiplication is not just a math fact to memorize.
It is a way to understand groups, patterns, totals, and strategy.

When children connect multiplication to sports, they can see that learning works like training:

Practice. Feedback. Adjustment. Growth.

A Brain Athlete does not become strong by getting everything right the first time.

A Brain Athlete becomes strong by continuing to train.

19. Brain Athlete Certificate

Certificate of Brain Athlete Training

This certifies that:

Name:

completed the QRU Sports Intelligence Lab(TM)
36 Sports Multiplication Challenges

and practiced:

- Multiplication
- Reasoning
- Focus
- Confidence
- Reflection
- Growth mindset

Coach Signature:

Date:

20. Final QRU Principle

Store understanding once.
Verify understanding once.
Manufacture unlimited expressions of understanding.

Continue your understanding at QRU: